# 6,906: Economics: Macro for Computer Science

## Lecture 4: Economic Growth and Technological Progress

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**March, 2026**

# Learning objectives

- By the end of this lecture, you should be able to:
  - Use one of the workhorse models of macroeconomics to analyze economic growth;
  - Appreciate the role of technological progress in driving cross-country differences in economic growth.

# What is a (macro) model?

- For an economist, a model is a simplified representation of the economy in which only the main ingredients are accounted for.
- Ideally, we would like to perform experiments as in other sciences, but this is possible only in a few areas of economics (behavioral economics). However, in macroeconomics things are different.
- We build macro models to conduct experiments that we cannot run in the real world and use the results from these experiments to inform policy-makers.
- A macroeconomic model is composed of a set of mathematical relationships through which one determines how variables affect each other (direction of relationships) and quantifies their impacts.

# The Solow model

- The Solow Model is the principal model for understanding long-run growth and cross-country income differences.
- It was developed by Robert Solow, for which he won a Nobel Prize!
- The key equations of the model are:
  1. an aggregate production function;
  2. a consumption/saving function;
  3. an accumulation equation for physical capital.

# The "micro world" of the Solow model

- Object: a closed economy.
- Time: A sequence of periods/years $t = 0, 1, 2, ...$
- Agents: Households and firms (and government)
- Commodities and markets: Output, capital services, and labor services.
- The market for output: Supply = firms' output, $Y_t$. Demand from households for consumption and investment $= C_t + I_t$ . The relative price equals 1.
- The market for capital services: Consumers own the capital stock $K_t$, and rent its services to firms. Relative price (in units of output) for renting one unit of capital for one period: $r_t$ = real rental rate for capital.
- Labor market: Households supply $= N_t =$ the labor force. Relative price: $w_t =$ the real wage rate.
- Competitive markets: $r_t$ and $w_t$ adjust to equate supply and demand in all markets - full (or natural) utilization of resources.

# The aggregate production function

- Denote $K_t$ as the capital stock and $N_t$ as the total number of workers in period $t$.
- Let $Y_t$ denote the output produced in period $t$. Suppose that there is a single representative firm that leases labor and capital from a single representative household each period to produce output. The production function is given by:

$$Y_t = F(K_t, N_t).$$

# The aggregate production function

The function $F(\cdot)$ is assumed to have the following properties:

- $F'_K > 0$ and $F'_N > 0$ (i.e. marginal products are always positive: more of either input means more output);
- $F''_{KK} < 0$, $F''_{NN} < 0$ (i.e. diminishing marginal products in both factors: more of one factor means more output, but the more of the factor you have, the less an additional unit of that factor adds to output);
- $F''_{KN} > 0$ (i.e. if you have more capital, the marginal product of labor is higher);
- $F(\lambda K_t, \lambda N_t) = \lambda F(K_t, N_t)$ (i.e. production function has constant returns to scale);
- $F(0, N_t) = F(K_t, 0) = 0$ (i.e. both capital and labor are necessary to produce).

# The aggregate production function

- The usual choice is the Cobb-Douglas production function: $F(K_t, N_t) = B(K_t)^{\alpha}(N_t)^{1-\alpha}$, where $\alpha$ is the capital share in income.
- The level of technology is measured by $B$, which stands for total factor productivity (TFP).
- The lack of time subscript $t$ on B is what makes this model the "basic" Solow Model.

# Problem of the firm

The optimization problem of the firm is to choose capital and labor to maximize profits ($\Pi_t$):

$$\max_{K_t, N_t} \Pi_t = F(K_t, N_t) - w_t N_t - r_t K_t,$$

- where $w_t$ denotes the real wage paid to workers and $r_t$ denotes the real return to capital.
- The first order conditions (FOCs) are:

$$w_t = F'_N = (1-\alpha) B (K_t)^{\alpha} N_t^{-\alpha}$$

$$r_t = F'_K = \alpha B (K_t)^{\alpha-1} (N_t)^{1-\alpha}$$

- FOCs imply that the firm should hire capital and labor to the point at which the marginal benefit of doing so (the marginal product of capital or labor) equals the marginal cost of doing so (the factor price).

# Problem of the household

- There exists a representative household in the economy. Its budget constraint is:

$$C_t + S_t = w_t N_t + r_t K_t + \Pi_t.$$

- Although separate decision-making units, the household owns the firm, hence dividend payments equal any profit earned by the firm.
- However, the firm does not make profit under the assumption of perfect competition ($w_t N_t + r_t K_t = Y_t$). In other words:

$$C_t + S_t = Y_t.$$

- "Biology":

$$N_{t+1} = (1 + n) N_t.$$

# Capital accumulation equation

- The Solow model assumes that savings represent a constant fraction of output:

$$S_t = sY_t,$$

  where $0 < s < 1$ denotes the saving rate (equivalently, the investment rate $S_t = I_t$).
- The initial level of capital, $K_t$, is given, but future levels of capital can be influenced through investment.

$$K_{t+1} = sY_t + (1 - \delta)K_t,$$

  where $0 < \delta < 1$ is the depreciation rate (the fraction of the capital stock that becomes obsolete each period).

# The complete set of equations

$$Y_t = B(K_t)^{\alpha}(N_t)^{1-\alpha}, \tag{1}$$

$$K_{t+1} = S_t + (1-\delta)K_t, \tag{2}$$

$$w_t = (1-\alpha)B(K_t)^{\alpha}N_t^{-\alpha}, \tag{3}$$

$$r_t = \alpha B(K_t)^{\alpha-1}(N_t)^{1-\alpha}, \tag{4}$$

$$C_t + S_t = Y_t, \tag{5}$$

$$S_t = sY_t, \tag{6}$$

$$N_{t+1} = (1+n)N_t. \tag{7}$$

5 Parameters: $B$, $\alpha$, $s$, $n$, and $\delta$. 7 Variables: $Y$, $K$, $N$, $r$, $w$, $S$, and $C$, of which $K_t$ and $N_t$ are state (predetermined) variables.

# Transformations in per worker terms

Define: $k_t = \frac{K_t}{N_t}$ and $y_t = \frac{Y_t}{N_t}$. Then:

$$\frac{K_{t+1}}{N_{t+1}} \frac{N_{t+1}}{N_t} = s \frac{Y_t}{N_t} + (1-\delta) \frac{K_t}{N_t},$$

and using that

$$N_{t+1} = (1+n) N_t,$$

we get

$$k_{t+1} = s \frac{y_t}{(1+n)} + (1-\delta) \frac{k_t}{(1+n)}.$$

# Transformations in per worker terms

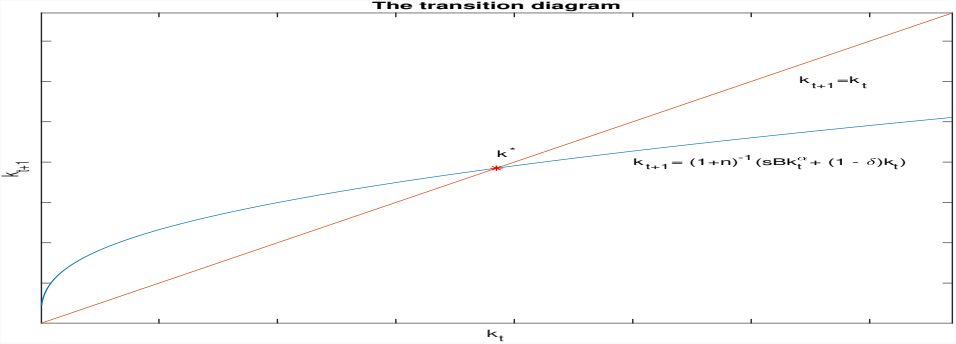
The set of equations with transformed variables in per worker terms is:

$$
\begin{aligned}
y_t &= B(k_t)^{\alpha}, \\
k_{t+1} &= s\frac{y_t}{(1+n)} + (1-\delta)\frac{k_t}{(1+n)}, \\
w_t &= (1-\alpha)B(k_t)^{\alpha}, \\
r_t &= \alpha B(k_t)^{\alpha-1}, \\
c_t + s_t &= y_t, \\
s_t &= sy_t.
\end{aligned}
$$

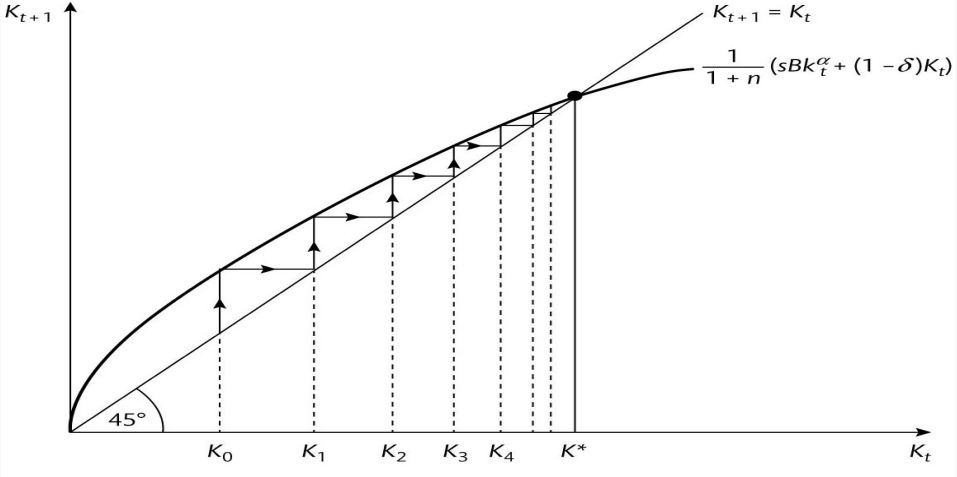
# The transition equation

Replacing the production function: $y_t = B(k_t)^\alpha$ in the capital accumulation equation, we obtain the transition equation:

$$k_{t+1} = s\frac{Bk_t^\alpha}{(1+n)} + (1-\delta)\frac{k_t}{(1+n)}. \tag{8}$$



# The convergence process




Source: Sorensen and Whitta-Jacobsen (2022)

# Slope of the transition equation

$$\frac{dk_{t+1}}{dk_t} = \frac{s\alpha B}{(1+n)} \frac{1}{k_t^{1-\alpha}} + \frac{(1-\delta)}{(1+n)}.$$

- If $k_t \to 0$ then the slope $\frac{dk_{t+1}}{dk_t} \to \infty$
- If $k_t \to \infty$ then the slope $\frac{dk_{t+1}}{dk_t} \to \frac{(1-\delta)}{(1+n)}$
- Interpretation: The plot starts out steep at the origin, and flattens out as $k_t$ gets bigger, eventually having a slope equal to $\frac{(1-\delta)}{(1+n)} < 1$. The 45 degree line has a slope of 1. The $k_{t+1}$ plot starts out with a slope greater than 1, but then it goes below 1. Since it is a continuous curve, this means that the plot of $k_{t+1}$ crosses the 45 degree line exactly once away from the origin. We indicate this point with $k^*$ and refer to it as the "steady state."

# Steady state

- The steady state is a natural point of interest. This is not because the economy is always at the steady state, but rather because, no matter where the economy starts (provided it does not start with $k_t = 0$), it will naturally gravitate towards this point.
- To algebraically solve for the steady state capital stock, take the transition equation (eq. 8) and set $k_{t+1} = k_t = k^*$:

$$k^* = B^{\frac{1}{(1-\alpha)}} \left( \frac{s}{n+\delta} \right)^{\frac{1}{(1-\alpha)}}$$

# Steady state

$$k^* = B^{\frac{1}{(1-\alpha)}} \left( \frac{s}{n+\delta} \right)^{\frac{1}{(1-\alpha)}},$$

$$y^* = B(k^*)^\alpha = B^{\frac{1}{(1-\alpha)}} \left( \frac{s}{n+\delta} \right)^{\frac{\alpha}{(1-\alpha)}},$$

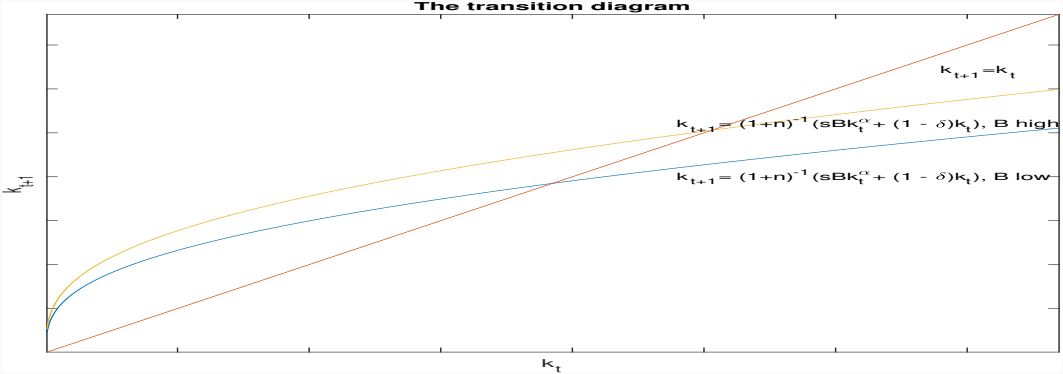
$$w^* = (1-\alpha)B(k^*)^\alpha = (1-\alpha)B^{\frac{1}{(1-\alpha)}} \left( \frac{s}{n+\delta} \right)^{\frac{\alpha}{(1-\alpha)}},$$

$$r^* = \alpha B(k^*)^{\alpha-1} = \alpha \left( \frac{s}{n+\delta} \right)^{-1},$$

$$c^* = (1-s)y^* = (1-s)B^{\frac{1}{(1-\alpha)}} \left( \frac{s}{n+\delta} \right)^{\frac{\alpha}{(1-\alpha)}},$$

$$s^* = sy^* = sB^{\frac{1}{(1-\alpha)}} \left( \frac{s}{n+\delta} \right)^{\frac{\alpha}{(1-\alpha)}}.$$

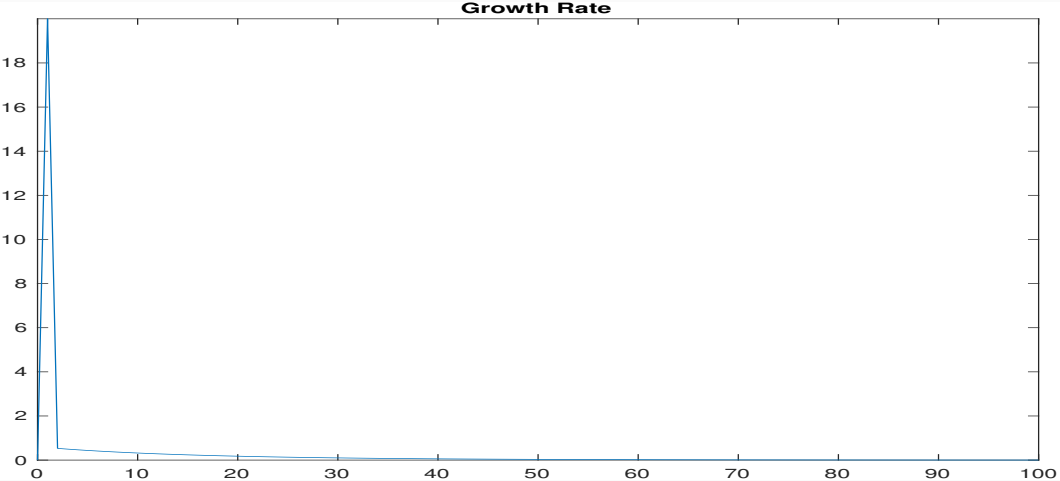
# Policy action: permanent increase in technology



# Policy action: permanent increase in technology

- $k^*$, $y^*$, $c^*$ increase if $B$ increases.
- Intuition: When the technology parameter $B$ shifts upwards, production per worker and hence savings per worker increase implying that now more capital is accumulated than needed to compensate for depreciation and population growth. As capital per worker increases, the marginal product of capital decreases until steady state, where savings per worker just cover the now increased 'replacement investment' needed to compensate for depreciation and population growth.

# Policy action: permanent increase in technology



# Limitations of the basic Solow model

- What is the growth rate of GDP per capita in steady state?
- Zero! Not in line with the stylized facts.
- However, there is transitory growth that is relatively long-lasting!
- In the Solow model, the transition towards the steady state is at least as important as the steady state itself, so the basic Solow model is a growth model!

# Conclusions of the basic Solow model

- What can a (poor) country do to create a transitory growth in GDP per worker resulting in a permanently higher level of income and consumption per worker?
- The basic Solow model provides the following answers:
  - Increase the savings rate
  - Reduce the growth rate of the labor force
  - Reduce the rate of depreciation, i.e., invest better
  - Improve the level of technology

# The general Solow model

- In the basic Solow model: no growth in GDP per worker in steady state.
- This contradicts the empirics for the Western world (stylized fact no. 5).
- In the general Solow model:
  - New variable, called labor augmenting productivity, $A_t$, is introduced and assumed to grow at a constant, exogenous rate (the only change).
  - This implies a steady state with balanced growth and a constant, positive growth rate of GDP per worker.
  - The source of long-run growth in GDP per worker in this model is exogenous technological growth.

# The general Solow model

- Same setup as in the basic Solow model.
- Only difference is the production function. Now there is the possibility of technological progress:

$$Y_t = B(K_t)^{\alpha}(A_t N_t)^{1-\alpha}.$$

- The full sequence $A_t$ is exogenous and $A_t > 0$ for all $t$.
- For simplicity, we can set $B = 1$:

$$Y_t = (K_t)^{\alpha}(A_t N_t)^{1-\alpha},$$

with the exogenous sequence, $A_t$, given by:

$$A_{t+1} = (1+g)A_t$$

# The complete set of equations

Assume: $F(K_t, N_t) = (K_t)^\alpha (A_t N_t)^{1-\alpha}$. Then:

$$
\begin{aligned}
Y_t &= (K_t)^\alpha (A_t N_t)^{1-\alpha}, \\
A_{t+1} &= (1+g)A_t, \\
K_{t+1} &= S_t + (1-\delta)K_t, \\
w_t &= (1-\alpha)(K_t)^\alpha N_t^{-\alpha} A_t^{1-\alpha}, \\
r_t &= \alpha (K_t)^{\alpha-1} (A_t N_t)^{1-\alpha}, \\
C_t + S_t &= Y_t, \\
S_t &= sY_t, \\
N_{t+1} &= (1+n)N_t.
\end{aligned}
$$

# Transformations in per worker terms

The set of equations with transformed variables in per worker terms is:

$$
\begin{aligned}
y_t &= (k_t)^{\alpha}(A_t)^{1-\alpha}, \\
k_{t+1} &= s\frac{y_t}{(1+n)} + (1-\delta)\frac{k_t}{(1+n)}, \\
w_t &= (1-\alpha)(k_t)^{\alpha}(A_t)^{1-\alpha}, \\
r_t &= \alpha(k_t)^{\alpha-1}(A_t)^{1-\alpha}, \\
c_t + s_t &= y_t, \\
A_{t+1} &= (1+g)A_t, \\
s_t &= sy_t.
\end{aligned}
$$

# Growth rate of output per worker

Let's take natural logs of the production function:

$$\begin{aligned} ln(y_t) &= ln((k_t)^{\alpha}(A_t)^{1-\alpha}) \\ &= \alpha ln(k_t) + (1-\alpha) ln(A_t) \end{aligned}$$

The growth rate of output is:

$$\begin{aligned} gr(y_t) &= \alpha gr(k_t) + (1-\alpha) gr(A_t) \\ gr(y_t) &= \alpha gr(k_t) + (1-\alpha) g \end{aligned}$$

Output and capital per worker grow at the same rate, which implies that $gr(y_t) = g$.

# Transformations in per 'Efficiency Unit of Labor' Terms

Define: $\tilde{k}_t = \frac{k_t}{A_t} = \frac{K_t}{A_t N_t}$ and $\tilde{y}_t = \frac{y_t}{A_t} = \frac{Y_t}{A_t N_t}$. Then:

$$\frac{K_{t+1}}{A_{t+1}N_{t+1}} = s\frac{Y_t}{N_t(1+n)A_t(1+g)} + (1-\delta)\frac{K_t}{N_t(1+n)A_t(1+g)},$$

which is equivalent to

$$\tilde{k}_{t+1} = s\frac{\tilde{y}_t}{(1+n)(1+g)} + (1-\delta)\frac{\tilde{k}_t}{(1+n)(1+g)}.$$

# Transformations in per 'Efficiency Unit of Labor' Terms

The set of equations with transformed variables in per efficiency unit of labor terms is:

$$
\begin{aligned}
&\tilde{y}_t = (\tilde{k}_t)^{\alpha}, \\
&\tilde{k}_{t+1} = s\frac{\tilde{y}_t}{(1+n)(1+g)} + (1-\delta)\frac{\tilde{k}_t}{(1+n)(1+g)}, \\
&\tilde{w}_t = (1-\alpha)(\tilde{k}_t)^{\alpha}, \\
&r_t = \alpha(\tilde{k}_t)^{\alpha-1}, \\
&\tilde{c}_t + \tilde{s}_t = \tilde{y}_t, \\
&\tilde{s}_t = s\tilde{y}_t.
\end{aligned}
$$

# The Transition Equation

Replacing the production function: $\tilde{y}_t = (\tilde{k}_t)^\alpha$ in the capital accumulation equation, we obtain the transition equation:

$$\tilde{k}_{t+1} = s\frac{(\tilde{k}_t)^\alpha}{(1+n)(1+g)} + (1-\delta)\frac{\tilde{k}_t}{(1+n)(1+g)}.$$

Set $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}^*$:

$$\tilde{k}^* = \left(\frac{s}{n+g+\delta+ng}\right)^{\frac{1}{(1-\alpha)}}$$

# Steady State

$$\tilde{k}^* = \left(\frac{s}{n+g+\delta+ng}\right)^{\frac{1}{(1-\alpha)}}$$

$$\tilde{y}^* = \left(\frac{s}{n+g+\delta+ng}\right)^{\frac{\alpha}{(1-\alpha)}}$$

$$\tilde{w}^* = (1-\alpha)\left(\frac{s}{n+g+\delta+ng}\right)^{\frac{\alpha}{(1-\alpha)}}$$

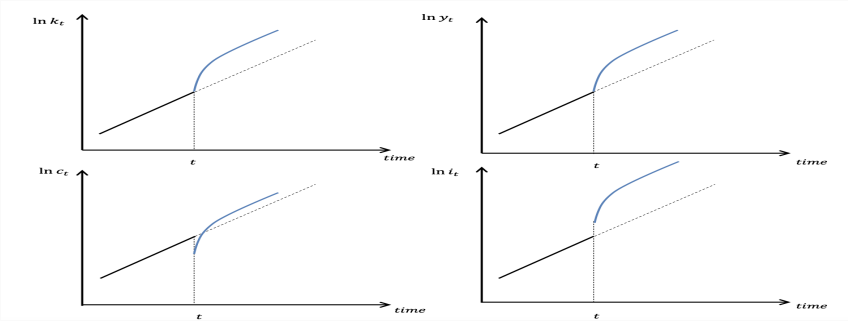
$$r^* = \alpha\left(\frac{s}{n+g+\delta+ng}\right)^{-1}$$

$$\tilde{c}^* = (1-s)\left(\frac{s}{n+g+\delta+ng}\right)^{\frac{\alpha}{(1-\alpha)}}$$

$$\tilde{s}^* = s\left(\frac{s}{n+g+\delta+ng}\right)^{\frac{\alpha}{(1-\alpha)}}$$

# Policy action: increase the savings rate $s$

- Prior to the increase in s, $y_t$, $k_t$, $c_t$, and $i_t$ would all be growing at rate $g$. Then, after the saving rate increase, these variables grow faster for a while. This puts them on a forever higher level trajectory, but eventually, the faster growth coming from more capital accumulation dissipates, and these variables grow at the same rate they would have in the absence of the increase in $s$.




Source: Garin, Lester, and Sims (2018)

# Policy action: increase the savings rate $s$

- By and large, countries are not poor because they lack capital or do not save enough - they are poor because they are unproductive. This means that policies which give these countries capital or try to increase their saving rates are not likely to deliver large changes in GDP per capita.
- Policies to lift these countries out of poverty need to focus on making these countries more productive, i.e. increase $g$.

# The determinants of technological progress

- Most technological progress is the outcome of firms' research and development (R&D) activities.
- The level of R&D spending depends not only on the fertility of research (how spending on R&D translates into new ideas and new products) but also on the appropriability of research results (the extent to which firms can benefit from the results of their own R&D).
- Patents give a firm that has discovered a new product the right to exclude anyone else from the production or use of that new product for some time.

# Policy action: increase productivity

- Factors which may contribute to overall productivity:
  - Knowledge and education: A more educated workforce is likely to coincide with more R&D.
  - Institutions: Laws that protect property rights and encourage innovation.
  - Finance: Relatively rich countries tend to have good financial institutions, which facilitates innovation.
  - Free trade: Countries with fewer barriers to international trade tend to have higher productivity. International trade in goods and services has two effects. First, like trade within a domestic economy, it allows for greater specialization. Second, trade in goods and services leads to knowledge spillovers from rich to poor countries, increasing the productivity levels in poor countries.

# Summary

- In this lecture we learned:
  - To use the Solow model to describe the stylized facts.
- Next lecture we will study the stylized facts of the short run.